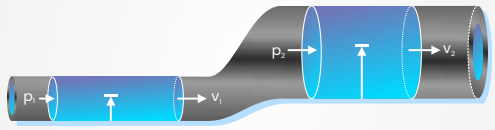
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KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Fluid Mechanics

(English Medium)

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FLUID MECHANICS

KEY CONCEPT

PART (A) - HYDROSTATICS

1. DENSITY AND RELATIVE DENSITY :

Density is mass per unit volume. Densities of solids and liquids are frequently compared with density of water and the relative density of a substance is defined as follows :

RELATIVE DENSITY with respect to water

$$= \frac{\text{density of the substance}}{\text{density of water}} = \frac{\text{mass of any volume of substance}}{\text{mass of an equal volume of water}}$$

(Also known as specific gravity of the substance) .

Note that relative density being a ratio of same type of physical quantities, it is a number only, without any physical dimension .

2. PRINCIPLE OF ARCHIMEDES :

The principle of Archimedes states that any body, totally or partially immersed in a fluid, experiences an upward force or thrust which is equal to the weight of fluid it displaces and acts vertically up through the C. G. of the displaced fluid. The term fluid covers liquids and gases .

3. LAW OF FLOTATION :

If a body floats in equilibrium in a fluid, its entire weight is supported by the upward thrust of the fluid. Hence, the weight of a floating body is equal to the weight of the fluid displaced by the body at the equilibrium state.

4. FLUID PRESSURE :

Pressure at any point in a fluid is defined as the normal force (or thrust) exerted by the liquid on the surface per unit area. Pressure is measured in dyne cm^{-2} in C G S units and in N m^{-2} in SI units, (also known as pascal) [Note that pressure is a scalar quantity].

When a plane surface is placed inside a liquid, the liquid exerts hydrostatic pressure on the surface, because of the weight of the liquid column above the surface. The total force exerted normally on the plane surface is called the thrust. The thrust over the surface is the vector sum of the thrusts over small area of the surface, over which the pressure can be considered to be uniform. Then:

(i) If the plane surface is horizontal, the pressure over the surface is uniform and the thrust = (area) × (the uniform pressure)

(ii) If the plane surface is rectangular with its plane vertical and a pair of sides horizontal, the thrust = (area) × (pressure at the centre of the area).

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5. PRESSURE IN LIQUIDS:

The hydrostatics pressure 'p' at any point in a liquid varies directly.

(i) As the vertical height (h) of the point below the surface &

(ii) As the density (d) of the liquid.

It can be shown that p = hdg.

When a liquid is at rest, the pressure is same at all points at the same horizontal level. The pressure at a point in a liquid does not depend on either the shape of the vessel or the area of cross-section of the vessel.

6. PASCAL'S LAW:

Pressure applied to a liquid (at rest) at one point is transmitted equally in all directions throughout the liquid. This is known as Pascal's Law.

The compressibility of all liquids is exceedingly small & for all practical purposes, liquids may be considered incompressible. Hence, the density is constant throughout the liquid.

Pressure at a point in a liquid is the same in all directions and is perpendicular to the surface upon which its acts.

7. GAUGE PRESSURE AND TOTAL PRESSURE (ABSOLUTE PRESSURE):

The total pressure at any point in a liquid is the addition of :

(i) The pressure due to the liquid above the point. It is called gauge pressure &

(ii) The atmospheric pressure acting on the surface of the liquid.

Thus total pressure = pressure due to liquid + atmospheric pressure

The total pressure is called absolute pressure.

8. THRUST DUE TO PRESSURE:

Total thrust on a horizontal surface immersed in a liquid = **(Pressure On Surface)** × **(Area Of Surface)**

Total thrust on a vertical surface immersed in a liquid = **(Pressure At C.G. Of Area)** × **(Area of Surface)**

PART (B) - HYDRODYNAMICS

1. The study of fluids in flow is called HYDRODYNAMICS.

2. VELOCITY - FIELD :

It is that space, where at every point in that space, there is a definite velocity. Then the space, where a fluid is in flow is a **Velocity-Field**.

3. FLOW - LINE :

In a velocity field is an imaginary line in that space, where the tangent to the line at any point on the line gives the direction of the velocity at that point. A flow line is also called a **Velocity-Line** or a **Stream-Line**.

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4. TYPES OF FLOW OF A FLUID :

Then the flow of a fluid can be classified as :

(i) A **Stream-Line Flow**, the stream lines in the flow space remains steady as time progresses.

(ii) A **Turbulent-Flow**, the stream lines in the flow space shift their positions as time progresses.

In a **Stream-Line Flow**, a group of stream lines form a tubular volume of the flow space, the surface of which is tangential to the stream lines, forming the lateral boundary of that tubular volume. Such a tubular volume in the flow space is a **Tube Of Flow**.

(iii) A steady state flow is the flow in which the fluid properties at any point in the velocity field do not change with time.

5. EQUATION OF CONTINUITY :

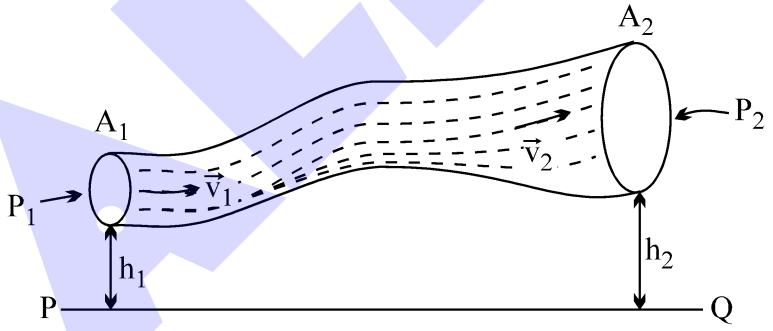
Equation of continuity states that for a steady state flow of a fluid in a pipe, the rate of mass flow across any cross section is constant.

$$\frac{dM}{dt} = \rho AV = \text{constant.}$$

If the fluid is incompressible density is constant at all points, hence, equation of continuity is AV = constant .

6. BERNOULLI'S EQUATION:

Consider a tube of flow in the space of the stream line flow of a fluid, in a uniform gravitational field. The flow is steady state.



Then : $\frac{P_1}{\rho}+\frac{V_1^2}{2}+gz_1=\frac{P_2}{\rho}+\frac{V_2^2}{2}+gz_2$

Generalizing and removing suffixes $\frac{P}{\rho}+\frac{V_1^2}{2}+gz = \text{constant.}$

This equation is called **Bernoulli's Equation** for steady, non-viscous and incompressible fluid flow.

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SURFACE TENSION

1. **Surface Tension :** Surface tension of a liquid is the normal force acting per unit length on either side of an imaginary line drawn the free surface of a liquid. The direction of this force is perpendicular to the line and tangential to the free surface of liquid. $T = \frac{F}{L}$

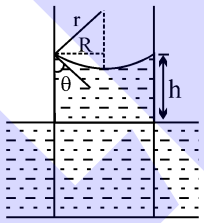
   Note: The surface tension of a liquid varies with temperature as well as dissolved impurities, etc. When soap mixed with water, the surface tension of water decrease.

2. **Surface Energy :** If the area of the liquid surface has to be increased work has to be done against the force of surface tension. The work done to form a film is stored as potential energy in the surface.
   $W = T \Delta A$

3. **Excess Pressure :** Excess pressure inside a liquid drop $\Delta p = \frac{2T}{r}$

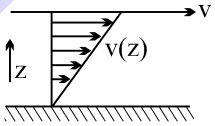
   For a soap bubble in air, there are two surfaces, and so, $\Delta p = 2 \times \frac{2T}{r} = \frac{4T}{r}$

4. **Capillarity :** Water in the capillary rises to a height $h = \frac{2T}{rg\rho}$ where r is the radius of meniscus, and r $= \frac{R}{\cos\theta}$ where θ is the angle of contact and thus $h = \frac{2T\cos\theta}{R\rho g}$



VISCOSITY

If a glass plate in contact with a water column of height h is moved with constant velocity v. Forces of viscosity appear between the solid surface and the layer in contact.



$$F = -\eta A \frac{dv}{dz}$$

where η is a constant called coefficient of viscosity, its CGS unit is poise.
Dimension is ML^1T^{-1}. The SI units of viscosity equal to 10 poise.

Stoke's Law and Terminal Velocity

When a sphere of radius r moves with a velocity v through a fluid of viscosity η, the viscous force opposing the motion of the sphere is $F = 6\pi\eta rv$

If for a sphere viscous force become equal to the net weight acting downward, the velocity of the body become constant and is known as terminal velocity.

$$6\pi\eta r v_T = \frac{4}{3}\pi r^3(\rho - \sigma)g \Rightarrow v_T = \frac{2}{9}r^2\left\{\frac{\rho-\sigma}{\eta}\right\}g$$

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